

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import urllib.request

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# -----
# INPUT IMAGE
# -----
url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/butterfly.jpg"
urllib.request.urlretrieve(url, "input.jpg")

img = cv2.imread("input.jpg")

if img is None:
    raise Exception("Image could not be loaded")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -----
# EXTRACT SIFT FEATURES
# -----
sift = cv2.SIFT_create()

keypoints, descriptors = sift.detectAndCompute(gray, None)

if descriptors is None:
    raise Exception("No descriptors detected")

print("Original descriptor shape:", descriptors.shape)

# Use at most 500 descriptors
descriptors = descriptors[:500]

# -----
# STANDARDIZATION
# -----
scaler = StandardScaler()
X = scaler.fit_transform(descriptors)

# -----
# PCA
# -----
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

print("\nPCA RESULT")
print("-" * 50)
print("Original dimensions :", X.shape[1])
print("Reduced dimensions  :", X_pca.shape[1])

print(
    "Variance ratio:",
    pca.explained_variance_ratio_
)

print(
    "Total variance retained:",
    sum(pca.explained_variance_ratio_) * 100,
    "%"
)

# -----
# VISUALIZATION BEFORE PCA
# -----
plt.figure(figsize=(8, 6))

plt.scatter(
    X[:, 0],
    X[:, 1],
    alpha=0.6
)

plt.xlabel("Original Feature Dimension 1")
plt.ylabel("Original Feature Dimension 2")
plt.title("Feature Representation Before PCA")
plt.grid(True)
plt.savefig("Q28_before_PCA.png")
plt.show()

# -----
# VISUALIZATION AFTER PCA
# -----
plt.figure(figsize=(8, 6))

plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    alpha=0.6
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Feature Representation After PCA")
plt.grid(True)
plt.savefig("Q28_after_PCA.png")
plt.show()

# -----
# CUMULATIVE VARIANCE
# -----
pca_full = PCA()
pca_full.fit(X)

cumulative = np.cumsum(
    pca_full.explained_variance_ratio_
)

plt.figure(figsize=(8, 5))

plt.plot(
    range(1, len(cumulative) + 1),
    cumulative,
    marker="."
)

plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Information Retention")
plt.grid(True)
plt.savefig("Q28_variance.png")
plt.show()
```

Original descriptor shape: (1124, 128)

PCA RESULT

Original dimensions : 128
Reduced dimensions : 2
Variance ratio: [0.10202645 0.07322779]
Total variance retained: 17.525423 %

